Chapter 18, Problem 33P

Problem

Find $f(t)$ if:

(a) $F(\omega) = 2 \sin \pi\omega[u(\omega + 1) - u(\omega - 1)]$

(b) $F(\omega) = \frac{1}{\omega}(\sin 2\omega - \sin \omega) + \frac{j}{\omega}(\cos 2\omega - \cos \omega)$

Step-by-step solution

Step 1 of 2

(A) $Let \quad x(t) = 2\sin \pi t \qquad [u(t+1)u(t-1)]$

From problem 17.9(b)

$$X(\omega) = \frac{4j\pi \sin\omega}{\pi^2 - \omega^2}$$

Applying Duality property,

$$f(t) = \frac{1}{2\pi} X(-t) = \frac{2j\sin(-t)}{\pi^2 - t^2}$$

$$f(t) = \frac{(2j\sin t)}{(t^2 - \pi^2)}$$

Step 2 of 2

(B) $F(\omega) = \frac{j}{\omega}(\cos 2\omega - j\sin 2\omega) - \frac{j}{\omega}(\cos\omega - j\sin\omega)$

$$= \frac{j}{\omega}\left(e^{j2\omega} - e^{-j\omega}\right)\frac{e^{-j\omega}}{j\omega} - \frac{e^{j2\omega}}{j\omega}$$

$$f(t) = \frac{1}{2}\operatorname{sgn}(t-1) - \frac{1}{2}\operatorname{sgn}(t-2)$$

$but \qquad \operatorname{sgn}(t) = 2u(t) - 1$

$$f(t) = u(t-1) - \frac{1}{2} - u(t-2) + \frac{1}{2}$$

$$= u(t-1) - u(t-2)$$